



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (E&C) (CBCS) Degree Examinations
 December 2023

EC2105-1 – NETWORK THEORY AND CONTROL SYSTEMS

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which law plays a significant role in the loop analysis of the network?
 A) KVL
 B) Law of Superposition Theorem
 C) KCL
 D) None of the above
- Ideal current source has
 A) Zero internal resistance
 B) Infinite internal resistance
 C) Low value of voltage
 D) Large value of current
- Nodal analysis is based on
 A) KVL
 B) Both
 C) KCL
 D) Law of conservation of energy
- A passive network is one which contains
 A) Only variable resistance
 B) Only some sources of e.m.f. in it
 C) Only two sources of e.m.f. in it
 D) No source of e.m.f. in it
- Thevenin's theorem applies to
 A) one port network
 B) three port network
 C) two port network
 D) four port network
- Application of Norton's theorem to a circuit yield
 A) Equivalent current source and impedance in series
 B) Equivalent current source and impedance in parallel
 C) Equivalent impedance
 D) Equivalent current source
- Superposition theorem can be applied only to circuits having
 A) Resistive elements
 B) Passive elements
 C) Nonlinear elements
 D) Linear bilateral elements
- For maximum transfer of power, internal resistance of the source should be
 A) Equal to load resistance
 B) Less than the load resistance
 C) Greater than the load resistance
 D) None of the above
- In determining short circuit admittance parameters, among V_1 , V_2 , I_1 , I_2 , which of the following are dependent variables?
 A) V_1 and V_2
 B) I_1 and I_2
 C) V_1 and I_2
 D) I_1 and V_2
- The hybrid parameter h_{11} is called?
 A) short circuit input impedance
 B) open circuit reverse voltage gain
 C) short circuit forward current gain
 D) open circuit output admittance
- Which of the following expression is true in case of open circuit impedance parameters?
 A) $V_1 = Z_{11} V_1 + Z_{12} I_2$
 B) $V_1 = Z_{11} I_1 + Z_{12} I_2$
 C) $V_1 = Z_{11} I_1 + Z_{12} V_2$
 D) $V_2 = Z_{11} I_1 + Z_{12} I_2$
- The net current through a port is
 A) Zero
 B) Infinity
 C) One
 D) None of the above
- The output of the feedback control system must be a function of
 A) Reference input
 B) Reference output
 C) Output and feedback signal
 D) Input and feedback signal

14. Routh Hurwitz criterion gives

- A) Number of roots in the right half of the s-plane B) Value of the roots
C) Number of roots in the left half of the s-plane D) Number of roots in the top half of the s-plane

15. Loop which does not possess any common node are said to be _____ loops.

- A) Forward gain B) Touching loops
C) Non touching loops D) Feedback gain

16. Which one of the following statements is not correct?

- A) Root loci can be used for analyzing stability and transient performance B) Shape of the root locus gives idea of type of controller needed to meet design specification
C) Root loci provide insight into system stability and performance D) Root locus can be used to handle more than one variable at a time

17. For relative stability of the system which of the following is sufficient?

- A) Gain margin B) Phase Margin
C) Both (a) and (b) D) None of these

18. Transfer function, when the bode diagram is plotted should be of the form

- A) $(1+T)$ B) $(1+S)$
C) TS D) $(1+TS)$

19. Gain margin of a first or second order system is

- A) 0 B) 10
C) 1 D) ∞

20. Cut off frequency is the frequency at which magnitude of closed loop frequency response is

- A) 3 db below its zero frequency B) 4 db below its zero frequency
C) 1 db below its zero frequency D) 2 db below its zero frequency

PART - B: DESCRIPTIVE ANSWER QUESTIONS**Unit – I**

Marks BT* CO* PO*

1. a) Reduce the circuit given below in Fig.1a into a single equivalent current source using source transformation.

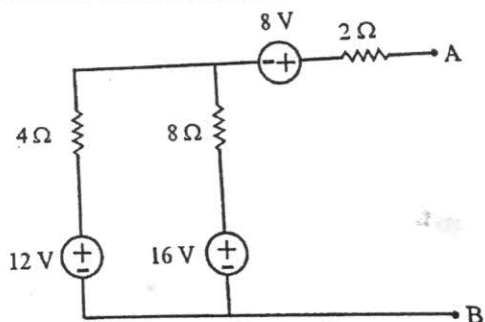


Fig. 1a

8 L*3 1 1

- b) Find the voltage across resistance R in the network shown in Fig. 1b using mesh analysis.

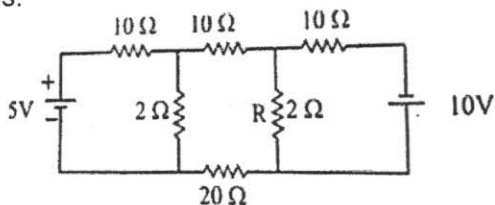
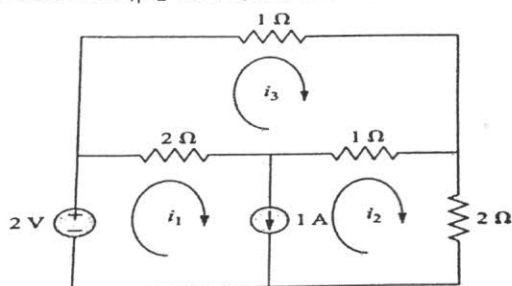


Fig. 1b

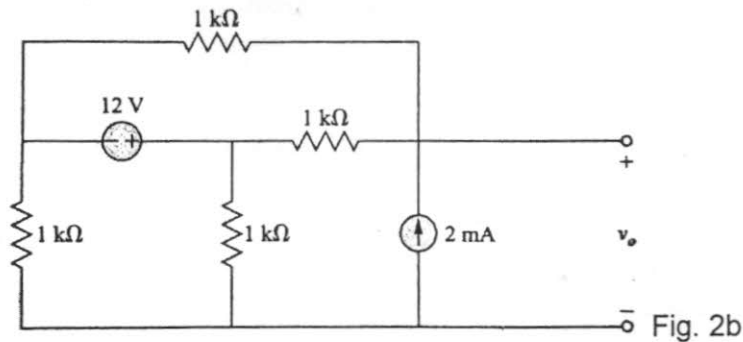
8 L3 1 1

2. a) Find the mesh currents i_1 , i_2 and i_3 for the network shown in Fig. 2a.

Fig. 2a
-2-

8 L3 1 1

- b) Use nodal analysis to find v_o for the network shown in Fig. 2b.



8 L3 1 1

3. a) Use superposition theorem to find i_o for the circuit shown in Fig. 3a.

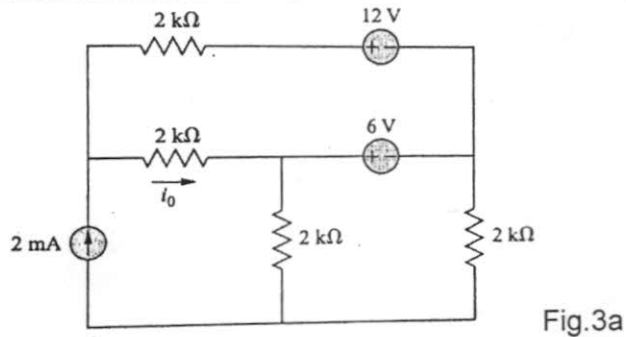


Fig.3a

8 L3 2 1

- b) Find the current through 16Ω resistor using Norton's theorem for the circuit shown in Fig. 3b.

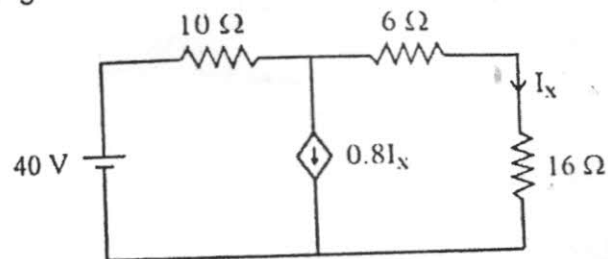


Fig. 3b

8 L3 2 1

Unit – II

4. a) Express open-circuit impedance parameters in terms of hybrid parameters.
b) Find the Z-parameters for the network shown in Fig. 4b.

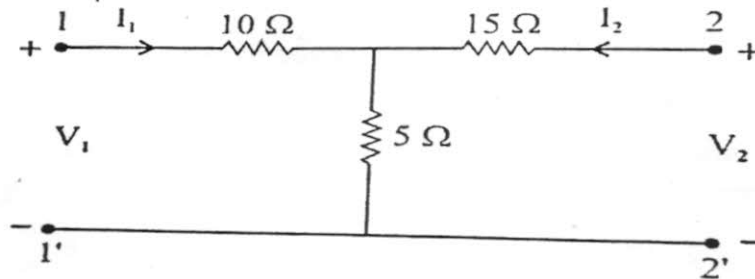


Fig. 4b

8 L3 3 1

5. a) List out the differences between open loop and closed loop control system.
b) Find $\frac{C}{R}$ for the graph shown in Fig. 5b using Mason's gain formula.

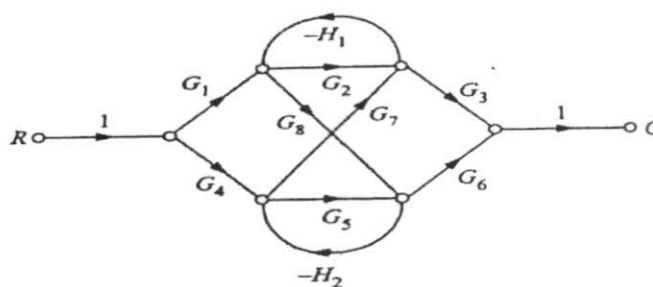


Fig. 5b

10 L3 4 1

6. a) Determine the range of K for stability of a unity feedback control system whose open loop transfer function is $G(s) = \frac{K}{s(s+1)(s+2)}$ 6 L3 4 1
- b) A negative feedback control system is characterized by $G(s)H(s) = \frac{K}{s(s+1)(s+2)(s+3)}$. Sketch the root locus plot for values of K ranging from 0 to ∞ . 10 L3 4 1

Unit – III

7. a) Construct Bode magnitude and phase diagrams for $GH(s) = \frac{100(0.1s+1)}{s(s+1)^2(0.01s+1)}$. Comment on the closed-loop stability of the system. 10 L3 5 1
- b) Determine the open loop transfer function of a system whose approximate plot is shown in Fig. 7b.

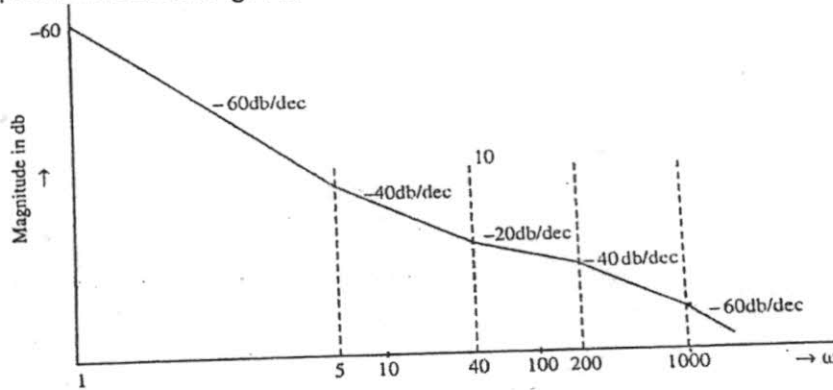


Fig. 7b

8. a) Investigate the stability of a negative feedback control system whose open-loop transfer function is given by $GH(j\omega) = \frac{50}{j\omega(0.5j\omega+1)(0.05j\omega+1)}$. Comment on the closed-loop stability of the system. 10 L3 5 1
- b) Determine the open loop transfer function of a system whose approximate plot is shown in Fig. 8b. (Note: -6db/octave is equivalent to -20db/decade)

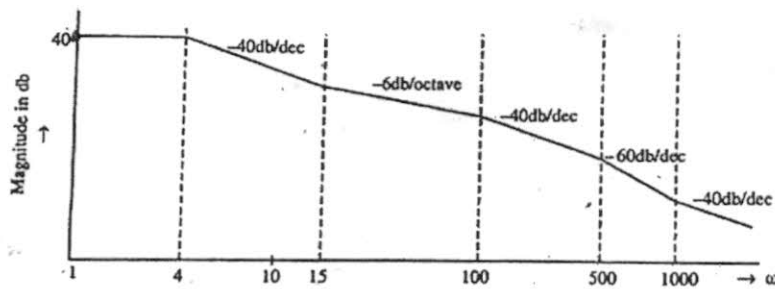


Fig. 8b

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
